NANO: Network Access Neutrality Observatory

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Georgia Tech

Net Neutrality

BusinessWeek

HOME INVESTING COMPANIES TECH INNOVATION MANAGING

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CEO Edward Whitacre talks about the AT&T Wireless acquisition and how he's moving to keep abreast of cable competitors

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Example: BitTorrent Blocking

It's Comcastic: Is Comcast Blocking Users From Seeding Torrents?

By Scott Gilbertson August 20, 2007 | 12:48:27 PM Categories: P2P

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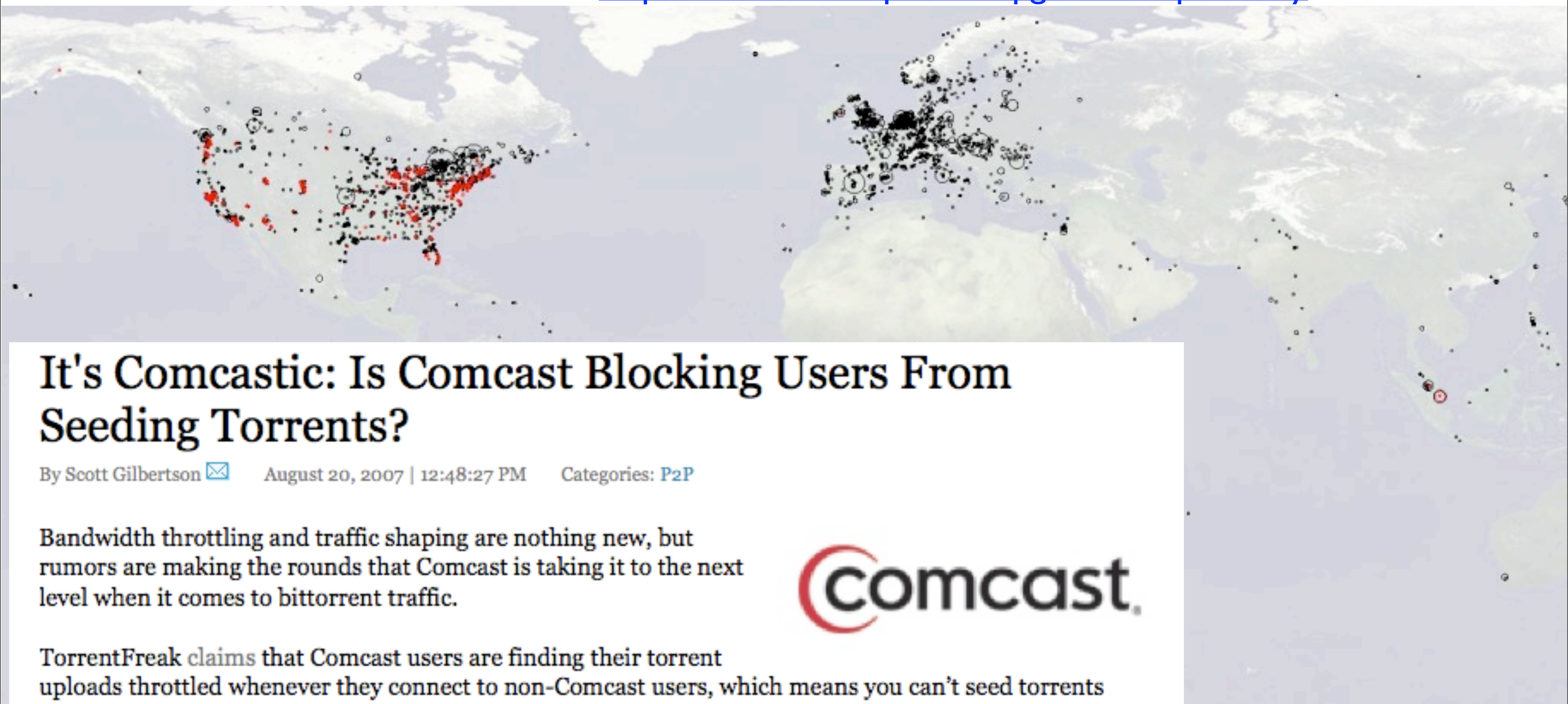
Example: BitTorrent Blocking

http://broadband.mpi-sws.mpg.de/transparenccy



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Many Forms of Discrimination

Throttling and prioritizing based on destination or service

Target domains, applications, or content

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- Throttling and prioritizing based on destination or service
    - Target domains, applications, or content
- Discriminatory peering
    - Resist peering with certain content providers
- ...

Problem Statement

Identify whether a degradation in a service performance is caused by discrimination by an ISP

Quantify the causal effect

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TCP RST (Glasnost)

ToS-bit based de-prioritization (NVLens)

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Quantify the causal effect

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Goal: Establish a *causal relationship* in the general case, without assuming anything about the ISP's methods

Causality: An Analogy from Health

- Epidemiology: study causal relationships between risk factors and health outcome
- NANO: infer causal relationship between ISP and service performance

Does Aspirin Make You Healthy?

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Sample of patients

Positive correlation in health and treatment

	Aspirin	No Aspirin
Healthy	40%	15%
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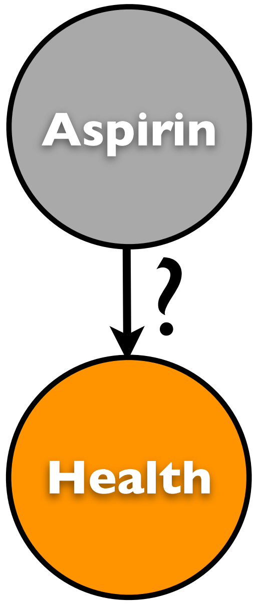

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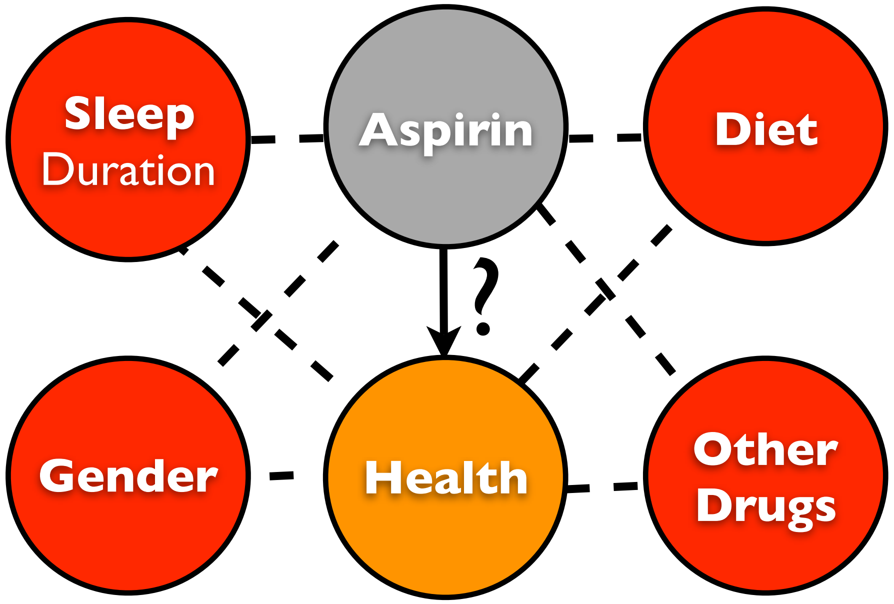

Confounding Variables: correlate with both cause and outcome variables and confuse the causal inference

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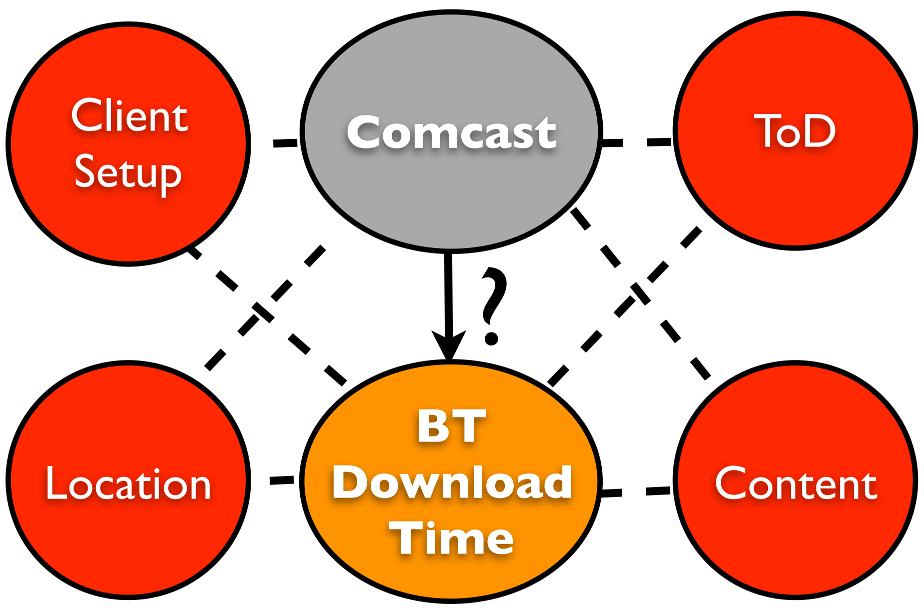

Many confounding variables can confuse the inference.

Causation vs. Association (1)

Causal Effect = E(Real Download time using Comcast) — E(Real Download time not using Comcast)

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G_1, G_0: Ground-truth values for performance
(aka. Counter-factual values)

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Problem: Generally, we do not observe both ground truth values for the same clients.
Consequently, *in situ* data sets are not sufficient to directly estimate causal effect.

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In general, $\alpha \neq \theta$.
How to estimate causal effect (θ) ?

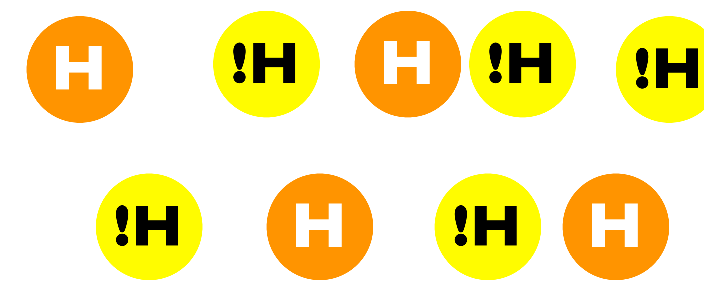
Estimating the Causal Effect

Two common approaches

a. Random Treatment

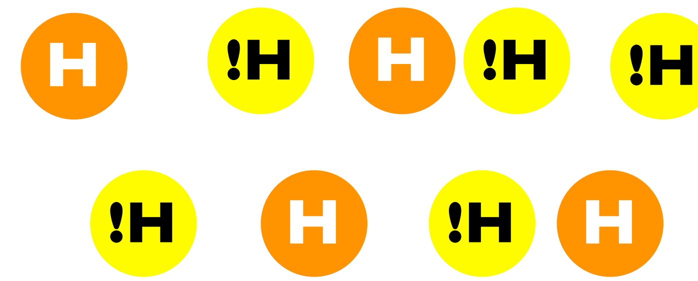
b. Adjusting for Confounding Variables

Random Treatment



Random Treatment

Given a population:



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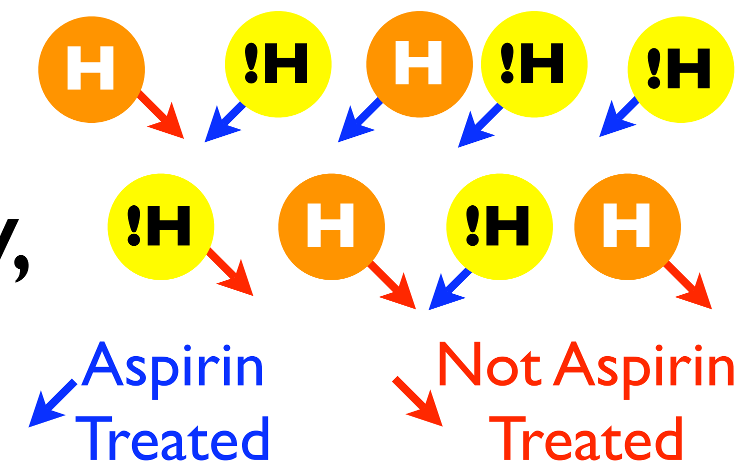
Given a population:

1. Treat subjects with Aspirin randomly, irrespective of their health



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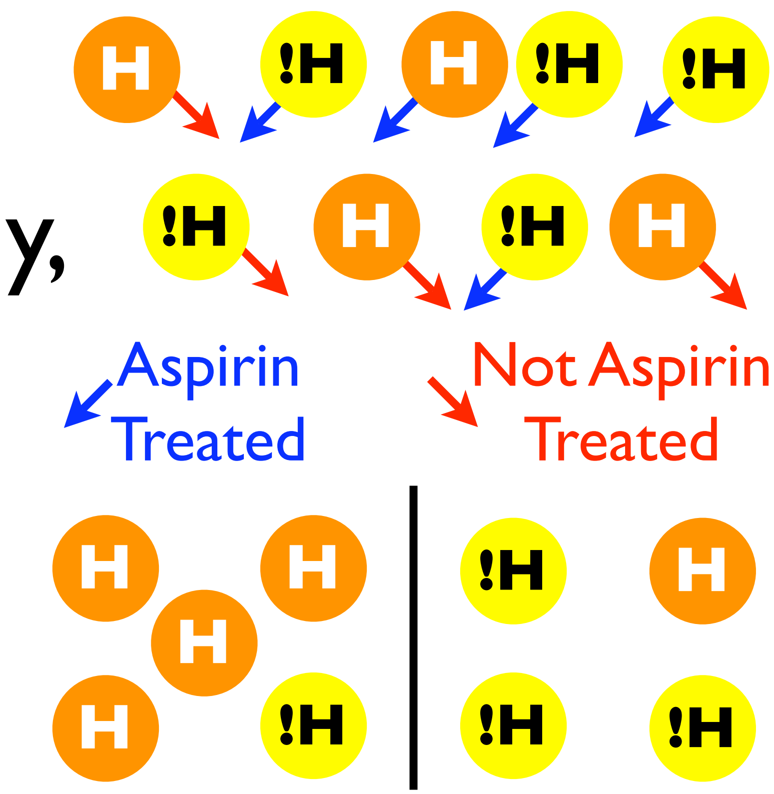


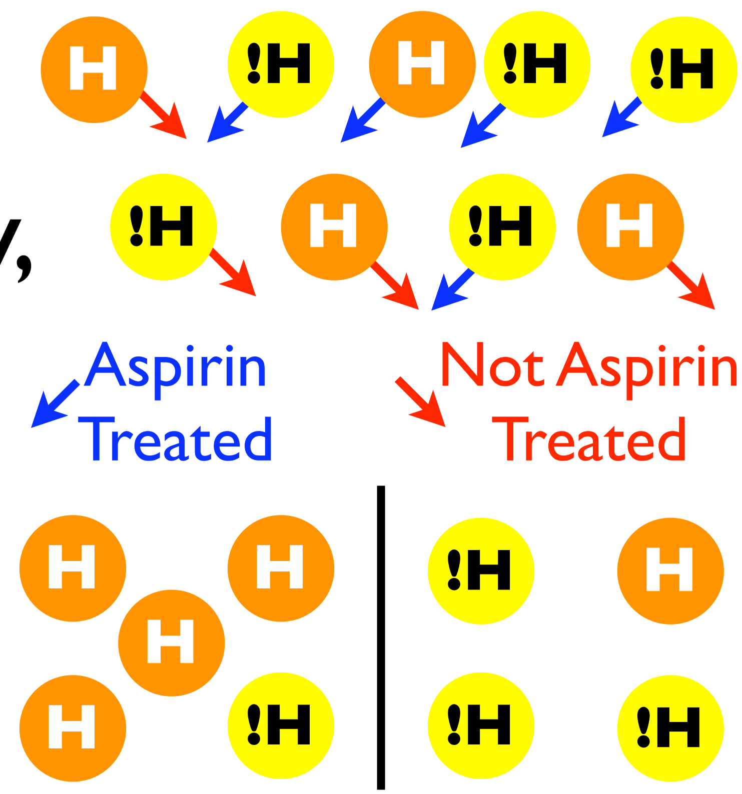
$\alpha = 0.8 - 0.25 = 0.55$

Random Treatment

Given a population:

1. Treat subjects with Aspirin randomly, irrespective of their health
2. Observe new outcome and measure association
3. For large samples, association converges to causal effect if confounding variables do not change

   Diet, other drugs, etc. should not change



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Random Treatment

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- Ask clients to change their ISP to an arbitrary one
- Difficult to achieve on the Internet

  Changing ISP is cumbersome for the users

  Changing ISP may change other confounding variables, i.e., the ISP network

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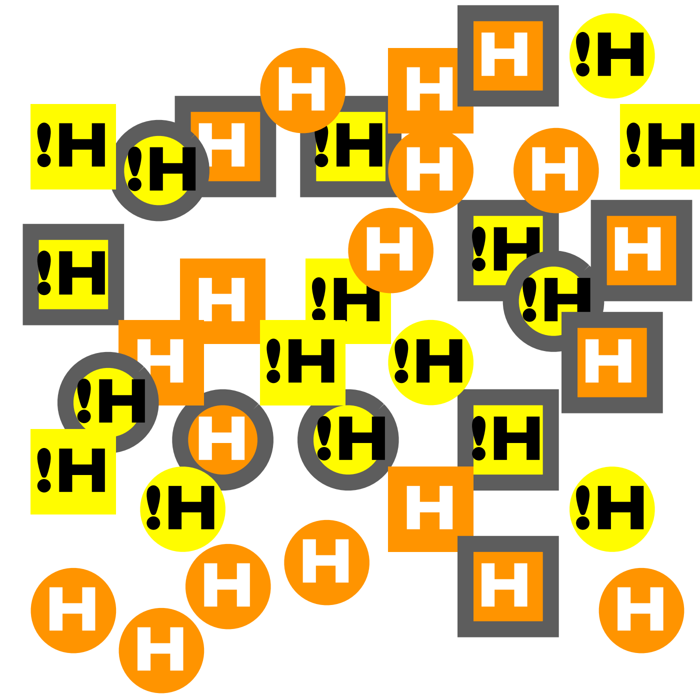


An *in situ* data set

Treatment:

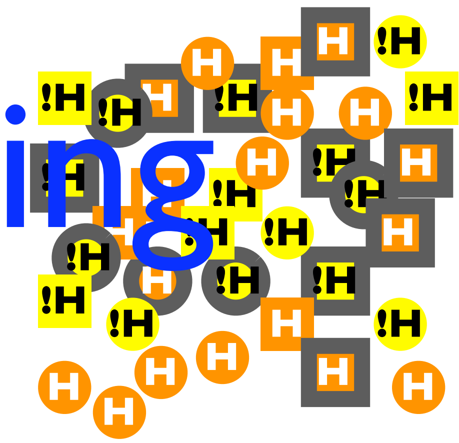
Baseline: Border(□,○)
Treated: No border

Outcome:

Healthy (H), Not Healthy (!H)

Stratum: Type {Circle, Square}

Adjusting for Confounding Variables



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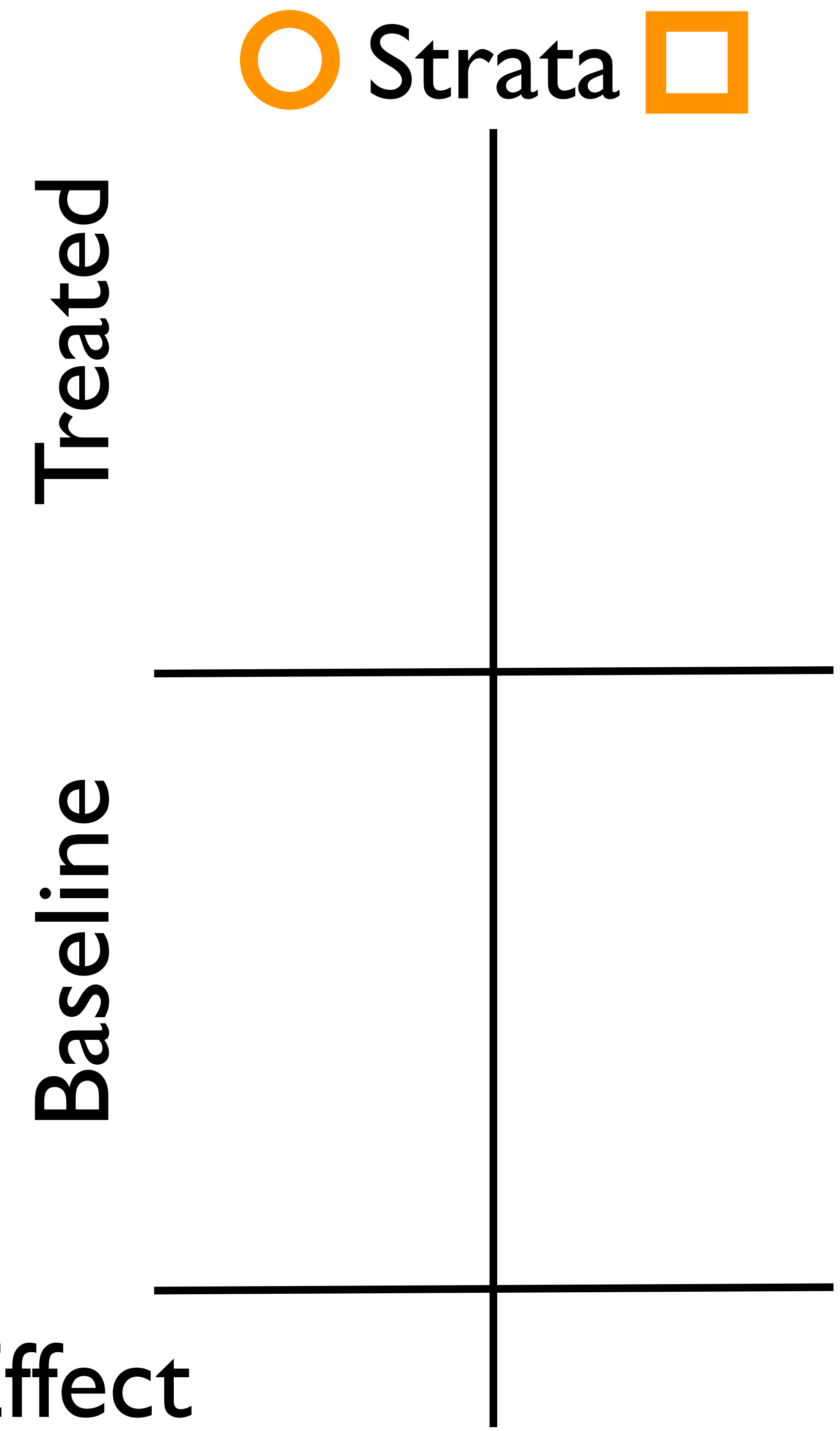

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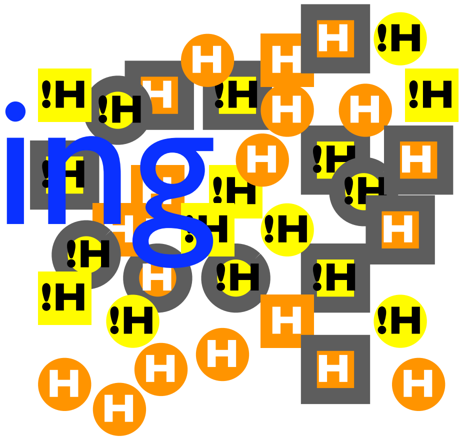

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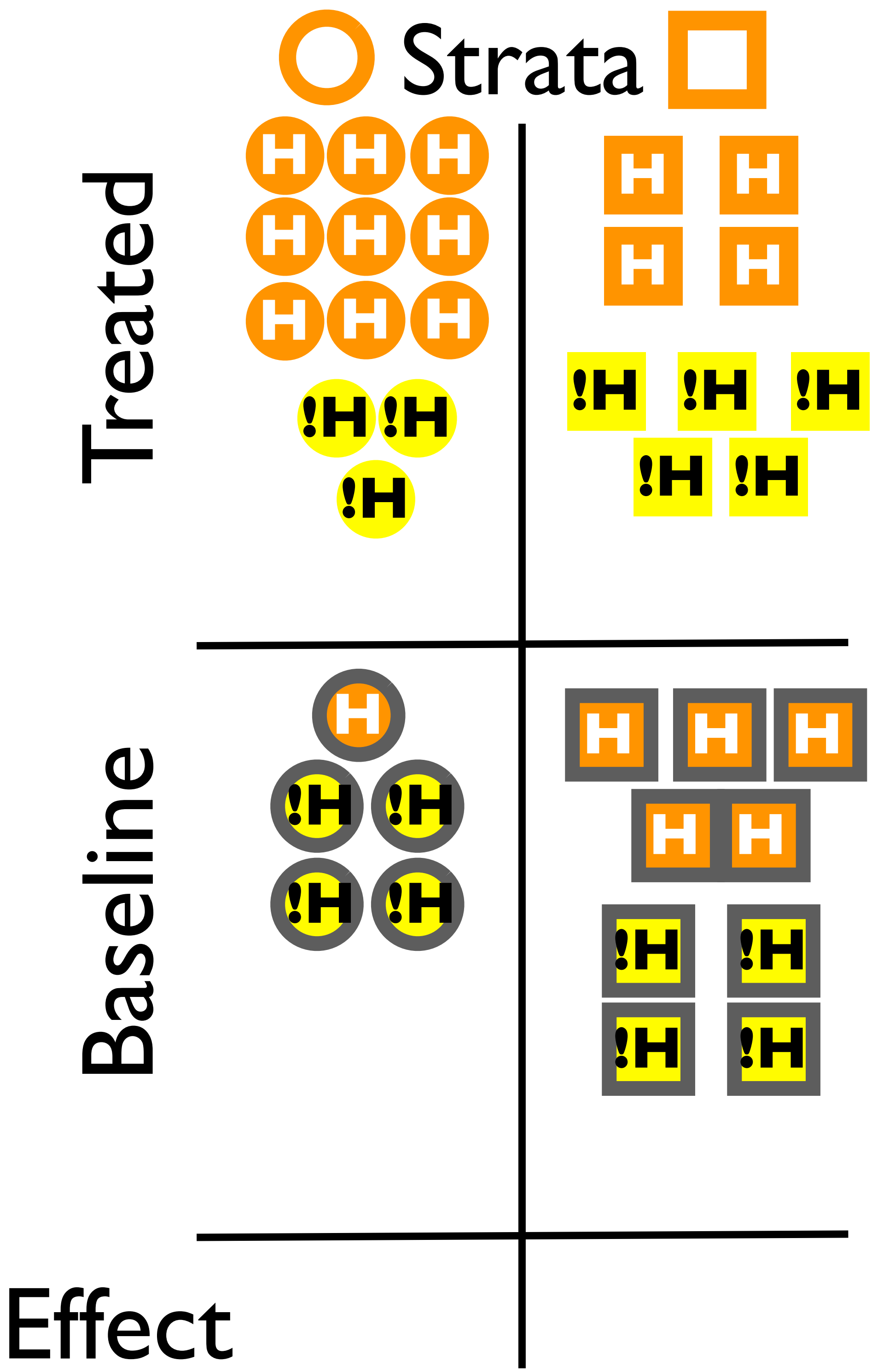

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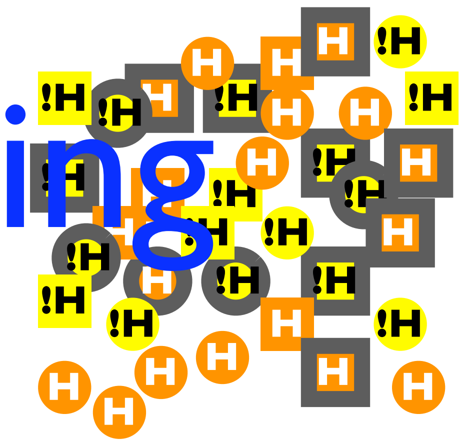

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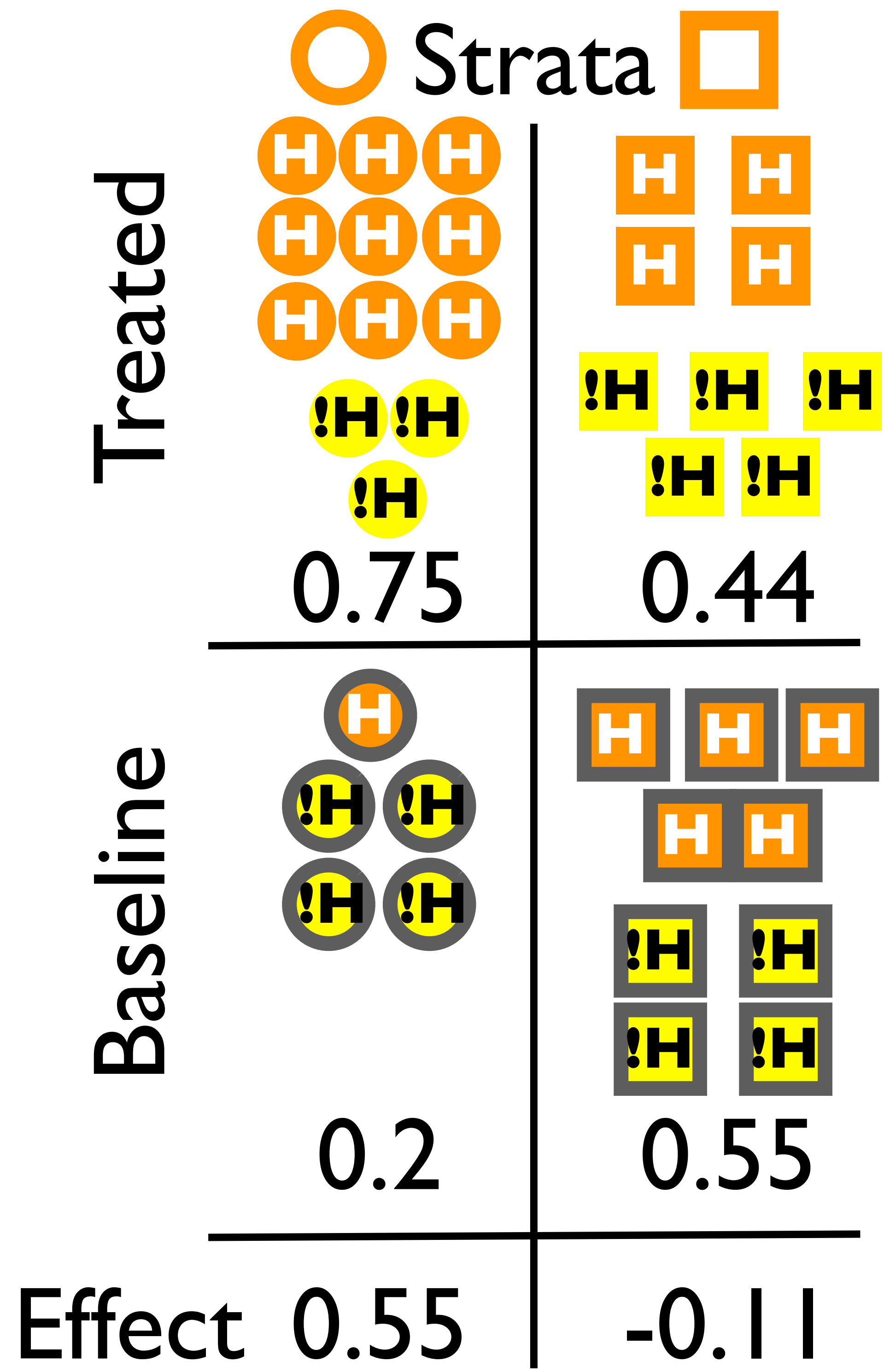

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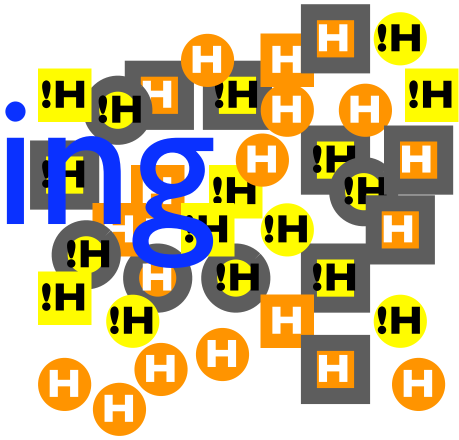

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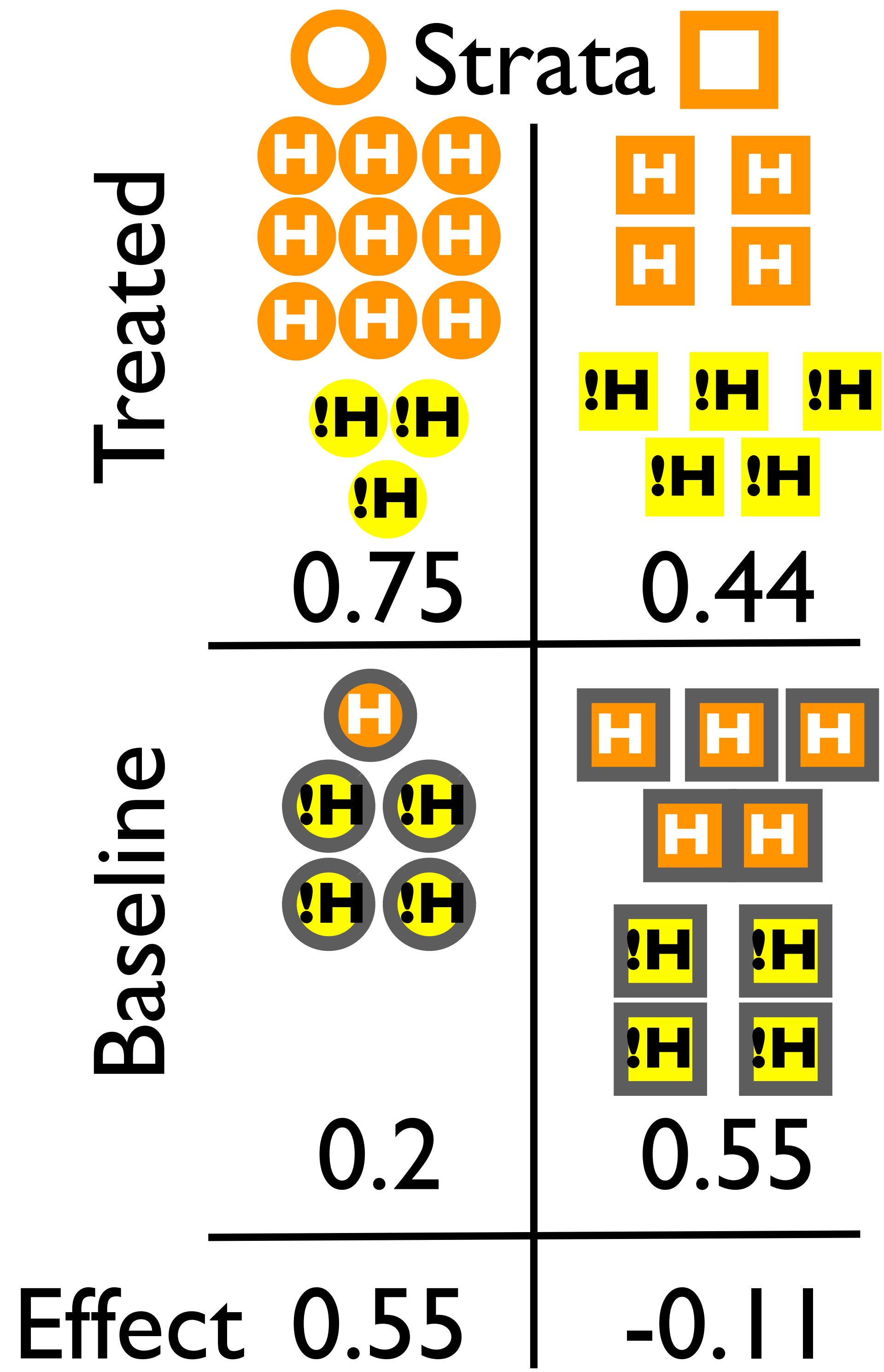
4. Measure Association
5. If there still is association, *then it must be causation*



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Can we infer more than the effect? e.g., the discrimination criteria

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c. Use service providers' model

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- ISP Related
  - Not all ISPs are equal; e.g., location.
- Temporal
  - Diurnal cycles, transient failures

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discriminated (-),
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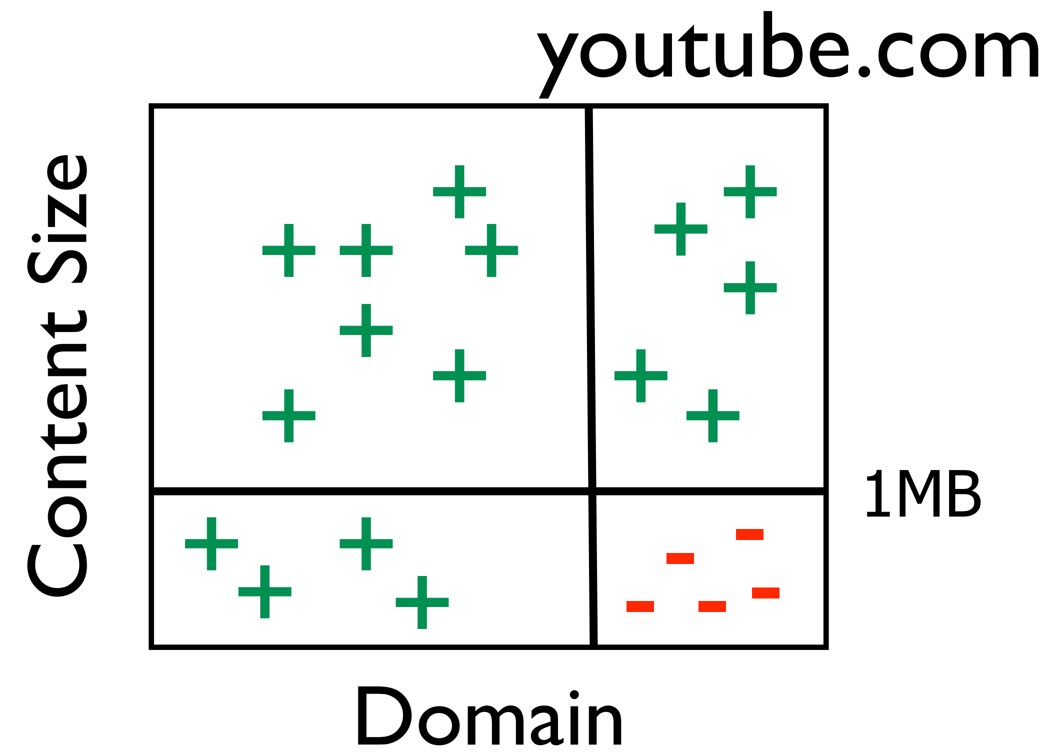

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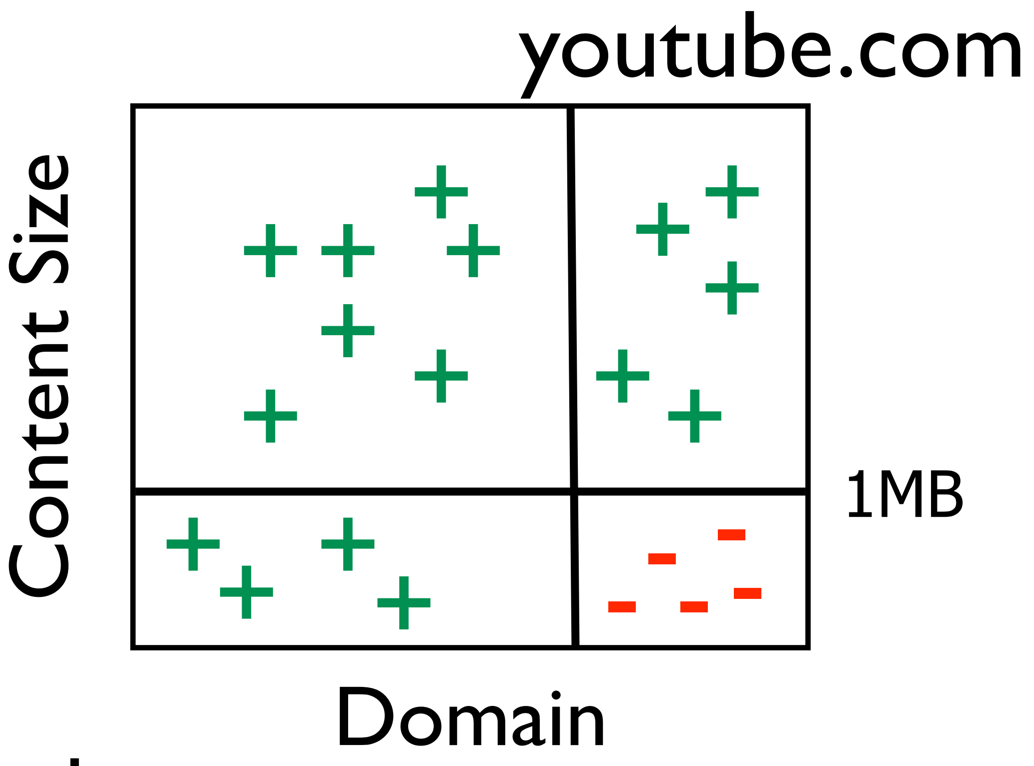
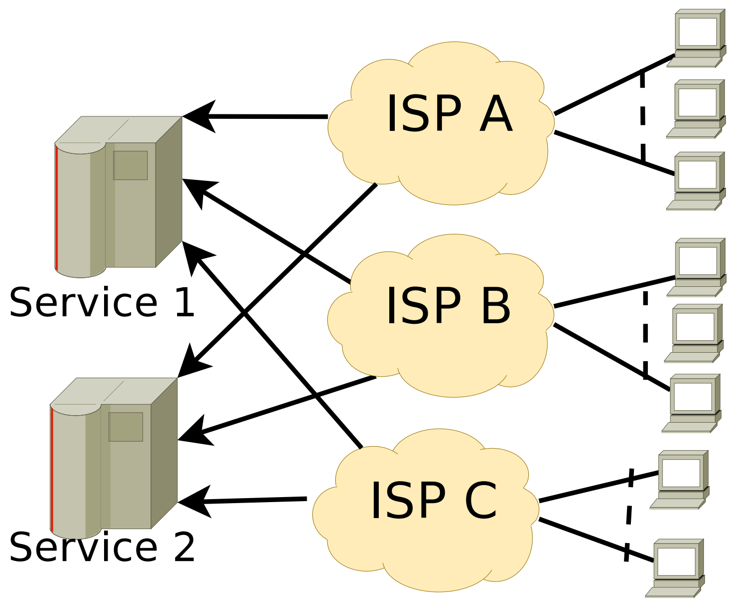


A Simple Simulation



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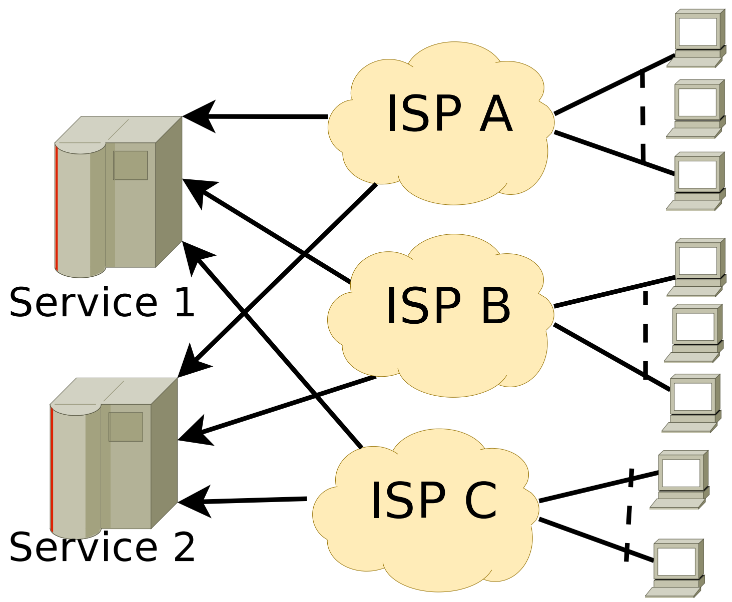

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- Service 1 is slower using App_2
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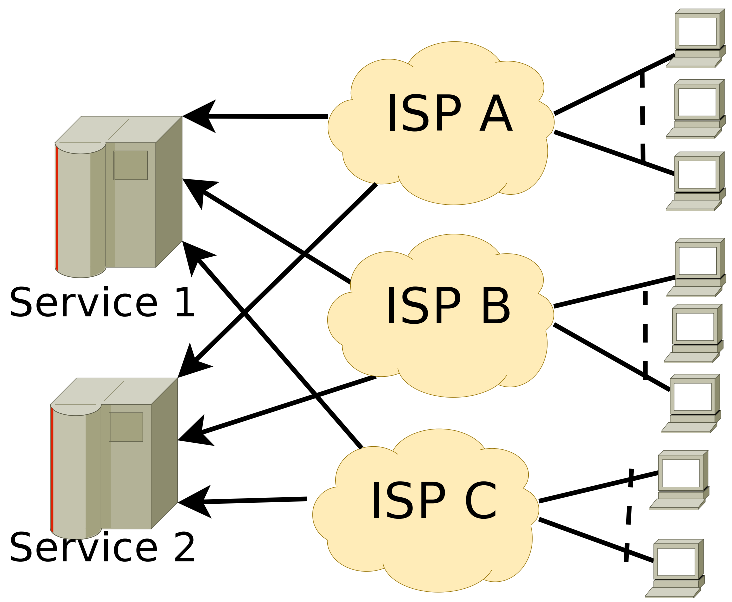

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Association	0.92 (10%)	0.04 (1%)

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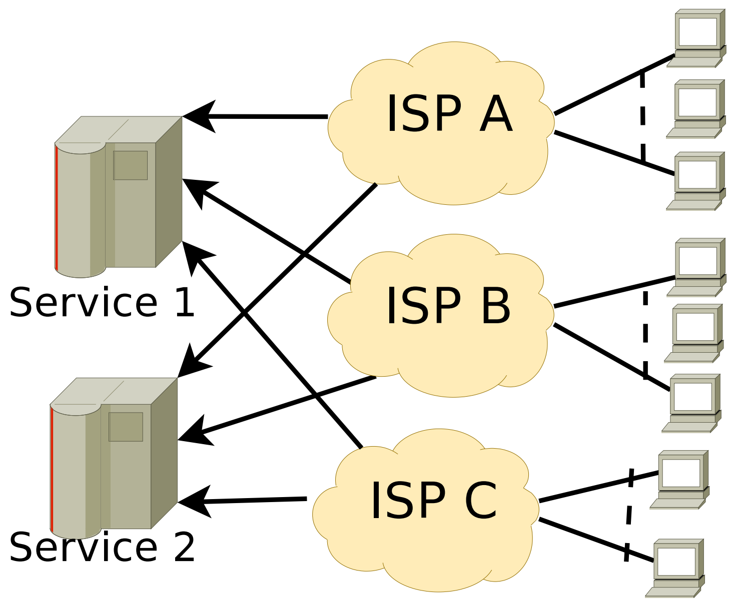

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Causation				
	Service 1		Service 2	
	App_1	App_2	App_1	App_2
Baseline	9.90	2.77	2.61	2.59
ISP_B	11.95	7.95	2.67	2.67
Causation	2.05 (20%)	5.18(187%)	0.06(2%)	0.12(4%)

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Contact: mmt@gatech.edu

Backup

Variables

- Causal Variables (X)

  The brand of an ISP

  IP address to ISP name mapping

- Outcome Variables (Y)

  Identify the Service

  The performance of a service

  Throughput, Delay, Jitter, Loss

- Confounding Variables (Z)

  Those that correlate with both

Adjusting for Confounding Variables

- Treatment (whether using ISP i): X_i
  Outcome (performance of service j): Y_j
  Causal Effect:

Adjusting for Confounding Variables

- Treatment (whether using ISP i): X_i
  Outcome (performance of service j): Y_j
  Causal Effect: $\theta_{ij}^{(z)} = \theta_{ij}(1;z) - \theta_{ij}(0;z)$

$$\theta_{ij} = \sum_z \theta_{ij}^{(z)}$$

$$\theta_{ij}(x;z) = \mathbb{E}(Y_j | X_i = x, Z = \mathbb{B}(z))$$

$\mathbb{B}(z)$ Boundaries of a stratum z with all-things equal

Sufficient Confounders?

If we have enough variables, then we should be able to predict the performance

Suppose: Confounding Var (Z), Treatment (X), Outcome (y)

Predict: $\hat{y} = f(X; Z)$

Test for prediction error: $|y - \hat{y}|/y$

Incentivizing

- Useful Diagnostics get a piggybacked ride

  NANO will identify transient failures in the elimination process

  Useful diagnostics dashboard for the users

P2P-*ized* Architecture

- Coordination to speed up inference and active learning